

Take-Home Task: Head of Data Technology

Objective:

The goal of this task is to assess your ability to propose and implement a suitable architecture in Snowflake for a streamlit application, the purpose of which is to provide internal teams with a downloadable excel reports, both an internal view (which reconciles to the ad platform and an external view which reconciles to what is billed to clients).

Context:

- There are two media data files:
 - A file extract from our Lumina media planning application
 - A download from Converge – our VOD demand-side / buying platform
 - Both datasets are linked by a common id called Vendor ID
 - Note: both these datasets are updated **daily** in the production system
- There is one config dataset: **advertiser_fee.csv** containing fees per advertiser. This is updated when contracts are novated or when new client acquired. The percentage fee should be applied to the spend on client facing reports.
- The **streamlit app** will be used for internal reporting, but the transformed data will also be sent to a) client-owned reporting platforms and b) other client-facing dashboards/datawarehouse solutions. This should dictate where business logic is applied.
- Updates to the app will be released every 2 weeks as new features are added.

Instructions:

- You can set up a free / trial snowflake account. This should have \$400 credit which is more than enough to complete task.
- Create an appropriate database structure for the data as provided considering good design principles.
- Although there is not much business logic/functionality implied in above I would like you to create a well structured streamlit app and python modules assuming that a lot more features and capabilities will be added. Please choose appropriate solution design patterns.
- Create a few simple automated tests for both the app and any python modules created showing how you would structure these as the application grows in complexity:
 - <https://docs.streamlit.io/develop/api-reference/app-testing>

Presentation

- Please forward your solution source code (db schema/DDDL + python files) in advance.
- As part of your presentation, you can do a code walk through, and feel free to create any simple diagrams needed to convey your
- Please also discuss how you would automate the building and deployment of your solution (i.e. CI/CD)